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Cooperative Occupancy Decision Making of Multi-UAV in Beyond-Visual-Range Air Combat: A Game Theory Approach

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ABSTRACT In this paper, a new cooperative occupancy decision-making problem for multiple unmanned aerial vehicles (UAVs) in beyond-visual-range (BVR) air combat is proposed; that is, UAV formation on each side first decides the occupancy positions of their own UAVs, then decides the target that each UAV will confront, so that each side can occupy the greatest predominance and be subject to the smallest threat possible. By analyzing the influence of the occupancy positions of the UAVs on the predominance gained by both sides and the threat against them, the problem is described as a zero-sum matrix game. Because the strategy combinations of this game are very numerous, as the best algorithm for the zero-sum matrix game, the Double Oracle (DO) algorithm still has difficulty solving the game model effectively. Therefore, we propose a DO combined with neighborhood search (DO-NS) algorithm to deal with the large-scale zero-sum matrix game. The results show that compared with the DO algorithm, the DO-NS algorithm can ensure the effectiveness of the computational time and solution quality.

INDEX TERMS Multiple UAVs cooperative occupancy decision making, BVR air combat, zero-sum matrix game, multiple UAVs cooperative target assignment, double oracle algorithm.

I. INTRODUCTION

With the development of missile technology and detection technology, beyond-visual-range (BVR) air combat has become the main mode of modern air combat [1], [2]. Compared with manned aerial vehicles, unmanned aerial vehicles (UAVs) are characterized by a lack of human casualties, high mobility, light weight and low cost, and these vehicles have become important members of modern air combat weapons and equipment [3]–[7]. Due to the limited number and performance of the weapons and sensors carried by a single UAV, its ability to carry out air combat tasks is limited. The effective coordination of multiple UAVs can better accomplish air combat tasks [8]–[10]. Recently, the cooperative BVR air combat decision making of multiple UAVs has attracted increasing attention [11]–[13].

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For the cooperative air combat decision making of multi-UAV over the horizon, most scholars are investigating the cooperative target assignment problem of multi-UAV, similarly to multi-aircraft cooperative target assignment problems, that is, determining how to effectively allocate the target for each UAV to confront on the premise of satisfying the constraints [14], [15]. For example, Su et al. calculated the overall predominance value of a UAV to a target based on four predominance factors (e.g., the velocity, the distance, the angle and the approaching time); then, a mixed integer linear programming model was established, and a self-organizing feature map-based optimization algorithm was proposed to find the optimal cooperative target assignment solution [14]. Based on Su's work, Hu et al. developed an improved ant colony optimization algorithm to solve the model [16]. In [15], Duan et al. calculated the utilities of UAV formation on both sides under each cooperative target assignment strategy combination based on the effectiveness and invalidity of UAV attacks on each other, and the problem

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was modeled as a bimatrix game that was solved by a particle swarm optimization and ant colony algorithm. Li et al. considered the kill probability of the missile carried by each aircraft to its target and the cost of the missile, formulated the problem as a mixed integer nonlinear programming model, and introduced an improved artificial fish swarm algorithm combined with an improved harmony search to solve the model [17]. Based on a mixed integer nonlinear programming model, Chang et al. calculated the overall predominance value of a UAV to a target from the perspective of the combat power potential energy [18]. Huang et al. simultaneously considered the survival probabilities of targets and targets threats to UAVs, an improved particle swarm optimization algorithm, an improved ant colony algorithm and an improved genetic algorithm were proposed to solve the model [19].

The multi-UAV cooperative target assignment described above is carried out on the premise that the positions of the UAVs are known to both sides. The positions of the UAVs on both sides are the key factors affecting the predominance of the UAVs to the targets, and these factors will have a great impact on the effect of the cooperative target assignment. Therefore, an effective occupancy strategy for the UAVs before the cooperative target assignment can further improve the expected predominance of multi-UAV in cooperative target assignment. For this reason, this paper focuses on a new cooperative occupancy decision making problem of multiple UAVs in BVR air combat. This problem is a two-stage decision making problem; that is, the decision makers first decide the occupancy position of each UAV, then decide the target that each UAV will confront, so that they can occupy the greatest predominance and be subject to the smallest threat possible. This problem is a kind of off-line decision making problem, which is insensitive to the computational time.

In this problem, when the occupancy positions of both sides' UAVs are determined, the two sides make decisions on the cooperative target assignments separately, and the maximum expected predominance values of both sides and the corresponding cooperative target assignment solutions can be subsequently calculated. At this point, the predominance of each side is the maximum expected predominance value of the cooperative target assignment, and the predominance of one side is the threat to the other side, while both sides of the confrontation hope that their own side has the greatest predominance and the smallest threat. Therefore, both sides need to consider not only their own occupancy strategies but also the opponent's occupancy strategies when making decisions. In addition, because each side of the confrontation cannot obtain the opponent's decision solution when making a decision, the decisions of both sides can be regarded as simultaneous decision making. In view of the above characteristics of the problem, this paper formulates the problem as a zero-sum matrix game and finds the mixed-strategy Nash equilibrium. The strategic action space of the game is the possible occupancy strategies for both sides in the first stage. The utility values of the strategy combinations are obtained by solving the cooperative target assignment problem under the corresponding occupancy strategies of both sides.

With an increasing quantity and occupancy positions of the UAVs, the strategies of the game increase sharply, which brings great difficulties to solving the game. Among the existing solution techniques for zero-sum matrix games, the Double Oracle (DO) algorithm enables solving a larger-scale zero-sum matrix game without a significant increase in computing time [20], [21]. This algorithm initially considers a restricted set of players' strategies and iteratively identifies the equilibrium to solve this restricted game; for each opponent's current optimal strategy, the player's best response strategy is identified within the entire strategy space and added to the restricted game, terminating after an iteration in which no new strategies are identified and the mixed-strategy Nash equilibrium of the game is found [22], [23]. However, when the scale of the game is very large, it takes a large amount of time to identify the best response strategy. Therefore, we propose a DO combined with neighborhood search (DO-NS) algorithm to solve the game. This algorithm solves the best response strategies by an iterative heuristic search, which can effectively reduce the search space and improve the efficiency of the algorithm. Experiments are designed to verify the effectiveness of the algorithm.

The main contributions of this paper are proposing a new cooperative occupancy decision-making problem for multiple UAVs in BVR air combat, formulating the problem as a zero-sum matrix game and designing an effective algorithm to solve the model. The rest of this paper is organized as follows. Section 2 introduces the mathematical description and modeling of the cooperative occupancy decision-making problem of multiple UAVs in BVR air combat and Section 3 describes the proposed DO-NS algorithm. Section 4 discusses and analyzes the experimental results. Finally, Section 5 concludes this paper with a short summary and suggestions for further research.

II. PROBLEM FORMULATION

In this section, we first abstract and mathematically describe the cooperative occupancy decision making problem of multi-UAV in BVR air combat. Second, we use a zero-sum matrix game to model the problem and define the mixed-strategy Nash equilibrium of the game model.

A. PROBLEM DESCRIPTION

UAV formations R and B confront each other in a three-dimensional region, denoted by $\Omega\left(\Omega \in \mathbb{R}^3\right)$, where Ω is a cubic region consisting of three parts, R and B are located on the two sides of Ω. As illustrated in Fig. 1(a), $\Omega_R\left(\Omega_R \in \mathbb{R}^3\right)$ is the occupancy area for R, $\Omega_B\left(\Omega_B \in \mathbb{R}^3\right)$ is the occupancy area for B, and $\Omega_O\left(\Omega_O \in \mathbb{R}^3\right)$ is the interval area for both sides. Due to the continuity of Ω_R and Ω_B, there are countless occupancy positions for the UAVs of R and B, and we divide the occupancy area into cubes of uniform size according to the spatial dimensions to make the cooperative occupancy

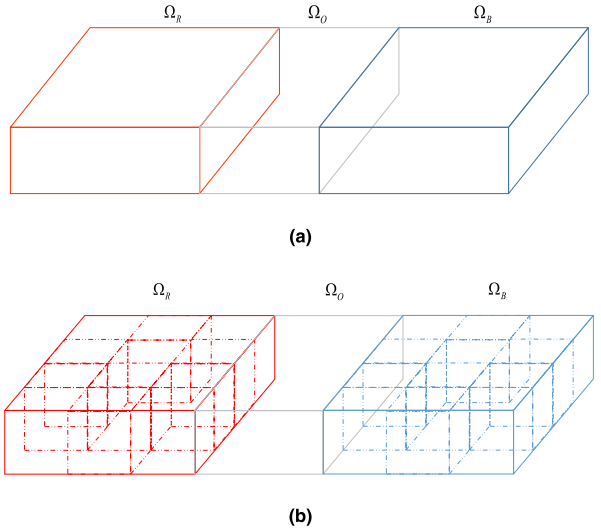


FIGURE 1. Abstraction and division of the occupancy areas of *R* and *B*. (a) Abstraction of the occupancy areas of *R* and *B*; (b) Division of the occupancy areas of *R* and *B*.

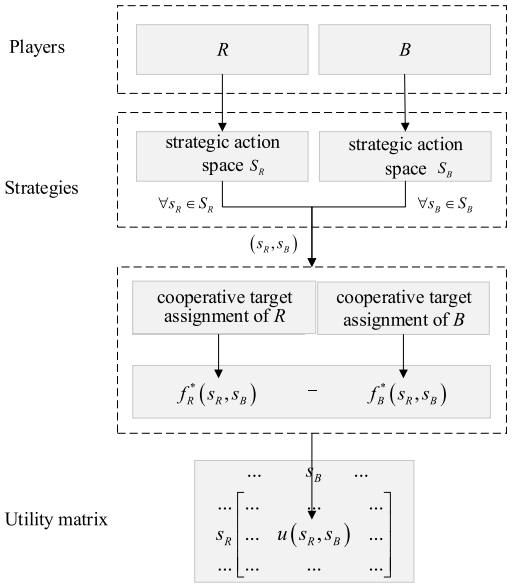


FIGURE 2. The framework of the zero-sum matrix game.

decision making of multiple UAVs a treatable problem (the subsequent cubes are referred to as discrete) [24], as illustrated in Fig. 1(b). At this point, the position of each UAV is the center of the cube that it occupies, and a cube can be used only as the occupancy position for a single UAV.

We denote the set of homogeneous UAVs of R as M, the set of homogeneous UAVs of B as N, the set of cubes of Ω_R as $C_R = \left\{c_R^i \,|i = 1, 2, \ldots, g\right\}$, and the set of cubes of Ω_B as $C_B = \left\{c_B^j \,|j = 1, 2, \ldots, g\right\}$. R and B first decide the cubes occupied by each UAV of their own side, and then, based on the occupancy positions of each UAV of the two sides, R and B decide the targets that each UAV of their own side will confront. The two-stage decision making is as follows.

In the first stage of decision making, R selects some cubes of C_R as the occupancy positions of its own UAVs, and B selects some cubes of C_B as the occupancy positions of its own UAVs. We denote this first-stage occupancy strategy of R as $\mathbf{d_R} = \left(d_R^e\right)_{e\in C_R}$, where $d_R^e \in \{0, 1\}$, and we require $\sum_{e\in C_R} d_R^e = |M|$. When a cube c_R^e is selected as the occupancy position of a UAV of R, $d_R^e = 1$; otherwise, $d_R^e = 0$. After the decision making, the UAVs in M are ranked according to the number of cubes that they occupy. For example, there are 2 UAVs in M that occupy the cube 4 and 8, the UAV occupies cube 4 is numbered UAV 1, and the UAV occupies cube 8 is numbered UAV 2. Similarly, we denote the first-stage occupancy strategy of B as $\mathbf{d_B} = \left(d_B^q\right)_{q\in C_B}$, where $d_B^q \in \{0, 1\}$ and $\sum_{q\in C_B} d_B^q = |N|$.

In the second stage of decision making, R decides on the UAVs assigned to each target on the basis of $\mathbf{d_R}$ and $\mathbf{d_B}$, and so does B. We denote the cooperative target assignment strategy of R as $\mathbf{a} = \left(a_{i,j}\right)_{i\in M, j\in N}$, where $a_{i,j} \in \{0, 1\}$, and when UAV i is assigned to the opponent's UAV j, $a_{i,j} = 1$; otherwise, $a_{i,j} = 0$. Similarly, we denote the cooperative target assignment strategy of B as $\mathbf{b} = \left(b_{j,i}\right)_{j\in N, i\in M}$, where $b_{j,i} \in \{0, 1\}$.

R makes the greatest predominance and the smallest threat by deciding on $\mathbf{d_R}$ and $\mathbf{a}$, and B by deciding on $\mathbf{d_B}$ and $\mathbf{b}$. We modeled the problem as a zero-sum matrix game, which is modeled in detail as follows.

B. GAME FORMULATION

In this section, the above cooperative occupancy decision-making problem for multi-UAV in BVR air combat is formulated as a zero-sum matrix game, where the occupancy strategies of R and B are regarded as the game strategies and the difference between the predominance and threat of R is the utility value under the combination of game strategies; R wants to maximize the expected utility values, and B wants to minimize the expected utility values. The framework of the model is shown in Fig. 2.

We formulate the above zero-sum matrix game model $G = (N, S, U)$ as follows:

- $N = R, B$ is the set of R and B players.
- $S = S_R \times S_B$ denotes the compact strategic action space for both players, where S_R and S_B are the available strategies for R and B, respectively. We denote a pure strategy of R as $s_R \in S_R$, and a pure strategy of B as $s_B \in S_B$, where $s_R \mapsto \mathbf{d_R}$ (i.e., the strategy s_R in the game theoretical context maps to a decision vector $\mathbf{d_R}$) and $s_B \mapsto \mathbf{d_B}$ (i.e., the strategy s_B in the game theoretical context maps to a decision vector $\mathbf{d_B}$). Let τ_R and τ_B denote the total number of possible strategies of R and B, and define $S_R = \left\{s_R^1, \ldots, s_R^i, \ldots, s_R^{\tau_R}\right\}$ and $S_B = \left\{s_B^1, \ldots, s_B^j, \ldots, s_B^{\tau_B}\right\}$, where τ_R and τ_B are the numbers of all possible cases for selecting $|M|$ and $|N|$ cubes from C_R and C_B, respectively. In addition, $|S_R| = \tau_R = \binom{g}{|M|}$, $|S_B| = \tau_B = \binom{g}{|N|}$, and $|S| = \tau_R\tau_B = \binom{g}{|M|}\binom{g}{|N|}$.

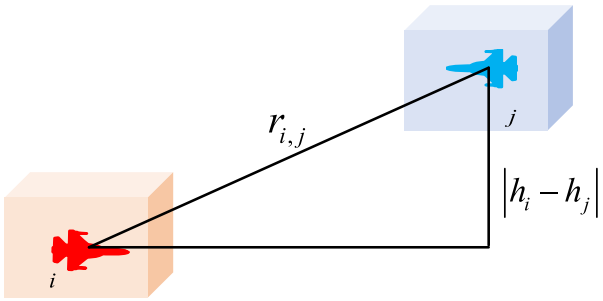


FIGURE 3. The relative positions of i and j in the occupancy area.

- $U = [u(s_R, s_B)]_{s_R \in S_R, s_B \in S_B}$ is the utility matrix of the game, where $u(s_R, s_B)$ is the utility of R under the strategy profile (s_R, s_B) and is obtained by calculating the difference in the predominance values of R and B in the second stage, that is, $u(s_R, s_B) = f_R^*(s_R, s_B) - f_B^*(s_R, s_B)$, and $f_R^*(s_R, s_B)$ and $f_B^*(s_R, s_B)$ are the maximum expected predominance values of the cooperative target assignment for R and B, respectively.

In matrix U, for each strategy profile (s_R, s_B), R and B make the second stage decision, and then, $u(s_R, s_B)$ is obtained. Based on Su's research work [14], the multi-aircraft cooperative target assignment problem is modeled as follows:

$$\text{MR} \qquad \max_{\mathbf{a}} f_R(s_R, s_B) = \sum_{i \in M} \sum_{j \in N} p_{i,j}^R \cdot a_{i,j} \tag{1}$$

$$s.t. \sum\nolimits_{j \in N} a_{i,j} = 1, \quad \forall i \in M \tag{2}$$

$$\sum\nolimits_{i \in M} a_{i,j} \le C_j, \quad \forall j \in N \tag{3}$$

$$\text{If } \exists i \text{ such that } \quad p_{i,j}^R \ge SF_R, \quad \text{then } a_{r,j} = 0 \text{ for } r \ne i \tag{4}$$

$$a_{i,j} \in \{0, 1\}, \quad \forall i \in M, \ \forall j \in N \tag{5}$$

The notation i represents the ith UAV in M, j represents the jth UAV in N, C_j is the maximum number of UAVs that can be assigned to UAV j, SF_R is the satisfaction of the predominance that is predefined by R, and $p_{i,j}^R$ is the predominance of i to j. Constraint (2) ensures that each UAV can be assigned to only one target UAV. Constraint (3) guarantees that the number of UAVs assigned to j does not exceed C_j. Constraint (4) avoids unnecessary assignment when j has already been assigned to one UAV with a predominance value larger than SF_R.

When solving the multi-UAV cooperative target assignment problem for R, we first need to calculate $p_{i,j}^R$. This paper mainly considers the distance predominance factor and height predominance factor that affect the predominance in BVR air combat, and the relative positions of i and j are shown in Fig. 3.

1) THE RELATIVE DISTANCE PREDOMINANCE

The relative distance predominance of i to j can be obtained by calculating the distance dominance function [14]

$$u_d(r_{i,j}) = \exp\left(\frac{-|r_{i,j} - r_i^m|}{2r_{i,j}}\right) \tag{6}$$

where

$$r_{i,j} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$
$$r_i^m = \frac{1}{2}\left(r_i^{\max} + r_i^{\min}\right)$$

The parameters (x_i, y_i, z_i) and (x_j, y_j, z_j) are the three-dimensional coordinates of i and j in the occupancy area, respectively. $r_{i,j}$ represents the relative distance between i and j, and $r_i^{\min}$ and $r_i^{\max}$ represent the minimum and maximum firing ranges of the weapons carried by i.

2) THE RELATIVE HEIGHT PREDOMINANCE

The relative height predominance of i to j can be obtained by calculating the height dominance function [25]

$$u_h(h_i, h_j) = \begin{cases} e^{-\frac{h_i - h_{ibest}}{h_{ibest}}} & h_{ibest} \le h_i \\ e^{\frac{h_i - h_{ibest}}{h_j}} & h_j < h_i \le h_{ibest} \\ -0.5 + \dfrac{h_i}{h_j} & h_i < h_j \end{cases} \tag{7}$$

where h_i and h_j are the heights of i and j, respectively, and h_{ibest} is the best air combat altitude of i.

The overall predominance degree of i over j can be computed as

$$p_{i,j}^R = w_1 u_d + w_2 u_h \tag{8}$$

where $w_1 + w_2 = 1$

The dominance value of each UAV in M over each UAV in N can be calculated from formula (8), and the maximum expected dominance value $f_R^*(s_R, s_B)$ under strategy profile (s_R, s_B) can be obtained by solving MR. By the same process, $f_B^*(s_R, s_B)$ can also be obtained.

C. MIXED-STRATEGY EQUILIBRIA

For noncooperative games, the Nash equilibrium is the most effective solution [26]. According to the nature of the zero-sum game, there is at least a mixed-strategy Nash equilibrium in the above game model G. Note that $\mathbf{x} \in \mathbf{X}$ is a mixed-strategy of R and that $\mathbf{x} = (x_1, x_2, \ldots, x_{\tau_R})$, where x_i is the probability placed on strategy s_R^i, and we require that $0 \le x_i \le 1$ and $\sum_{i=1}^{\tau_R} x_i = 1$. Note that $\mathbf{y} \in \mathbf{Y}$ is a mixed-strategy of B and that $\mathbf{y} = (y_1, y_2, \ldots, y_{\tau_B})$, where y_j is the probability placed on strategy s_B^j, and we require that $0 \le y_j \le 1$ and $\sum_{j=1}^{\tau_B} y_j = 1$. Under a mixed-strategy profile $(\mathbf{x}, \mathbf{y})$, the expected utility of R is $F(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{\tau_R} x_i \sum_{j=1}^{\tau_B} y_j u\left(s_R^i, s_B^j\right)$, and the expected utility of B is $-F(\mathbf{x}, \mathbf{y})$. If there is a mixed-strategy profile $(\mathbf{x}^*, \mathbf{y}^*)$ that satisfies

$$F(\mathbf{x}, \mathbf{y}^*) \le F(\mathbf{x}^*, \mathbf{y}^*) \le F(\mathbf{x}^*, \mathbf{y}) \tag{9}$$

then, the mixed-strategy profile $(\mathbf{x}^*, \mathbf{y}^*)$ is a Nash equilibrium of game G [27].

According to the properties of the mixed-strategy Nash equilibrium in [28], inequality (9) can be formulated by the following expression, and the corresponding optimal solution of formulation (10) is the mixed-strategy Nash equilibrium of game G:

$$\max_{\mathbf{x}} \min_{\mathbf{y}} \sum_{i=1}^{\tau_R} x_i \sum_{j=1}^{\tau_B} y_j u\left(s_R^i, s_B^j\right) = \min_{\mathbf{y}} \max_{\mathbf{x}} \sum_{i=1}^{\tau_R} x_i \sum_{j=1}^{\tau_B} y_j u\left(s_R^i, s_B^j\right) \quad (10)$$

III. DO-NS ALGORITHM

The scale of the game presents a combined explosion with increases in the number of cubes in the occupied area and the number of UAVs on both sides. We introduce the neighborhood search technology and propose the DO-NS algorithm. The DO-NS algorithm can obtain the best response strategies by an iterative heuristic search, which can effectively reduce the search space. The framework of the proposed DO-NS algorithm is shown in Fig. 4.

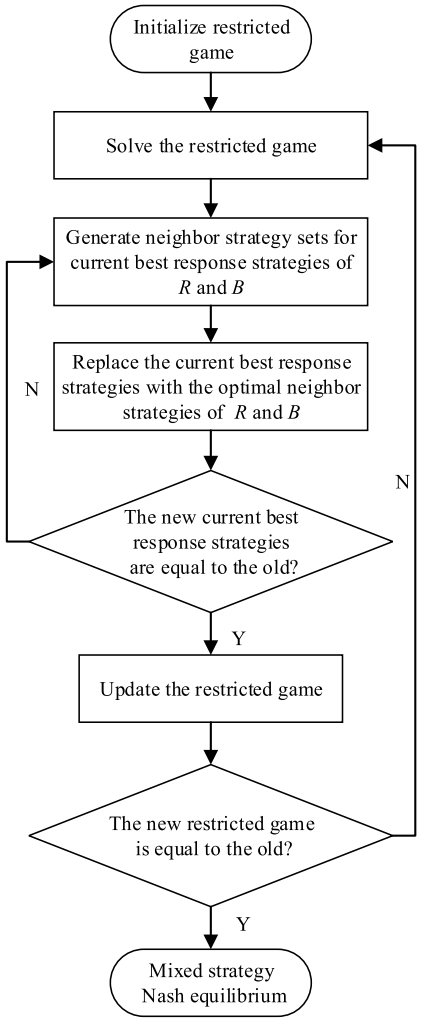


FIGURE 4. The framework of the proposed DO-NS algorithm.

In the implementation of the DO-NS algorithm, similar to the DO algorithm designed by Han *et al.* [29], we use four interrelated algorithmic procedures to solve the mixed-strategy Nash equilibrium of game G: AttackLP, CoreLP, Red Oracle and Blue Oracle. The pseudocode of the DO-NS algorithm is shown in algorithm 1.

Algorithm 1 DO-NS Algorithm

1. Initialize S_R^0 as a subset of one or more pure strategies of R;
2. Initialize S_B^0 as a subset of one or more pure strategies of B;
3. k = 0;
4. repeat
5. $U^k \leftarrow$ AttackLP (S_R^k, S_B^k)
6. $(\hat{\mathbf{x}}, \hat{\mathbf{y}}) \leftarrow$CoreLP$(G^k)$
7. $\bar{s}_R \leftarrow$Red Oracle$(\hat{\mathbf{x}}, \hat{\mathbf{y}})$
8. $\bar{s}_B \leftarrow$Blue Oracle$(\hat{\mathbf{x}}, \hat{\mathbf{y}})$
9. $S_R^{k+1} \leftarrow S_R^k \cup \bar{s}_R$
10. $S_B^{k+1} \leftarrow S_B^k \cup \bar{s}_B$
11. $k \leftarrow k+1$
12. until $S_R^k = S_R^{k-1}$ and $S_B^k = S_B^{k-1}$
13. return $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$

The DO-NS algorithm first needs to consider a smaller set of pure strategies of R and B as the strategy set of restricted game G^0 in the 0th iteration, and we note this set as tuple $S^0 = (S_R^0 \times S_B^0)$ (lines 1-2). In the kth iteration, the set of strategies of G^k is $S^k = (S_R^k \times S_B^k)$. Then AttackLP (Section II-B) is used to solve the multi-UAV cooperative target assignment problem for each strategy profile in S^k, and the utility value of each strategy profile is obtained. Then, the restricted game matrix U^k is obtained (line 5). Next, the corresponding mixed-strategy Nash equilibrium $(\hat{\mathbf{x}}, \hat{y})$ of restricted game G^k is solved by CoreLP (Section III-A) (line 6). Then, Red Oracle and Blue Oracle (Section III-B) are used to identify the best response strategy of R or B given a current mixed-strategy of the opponent (lines 7-8), S^k is updated, and then, $k = k + 1$. The process repeats until no new best response strategy can be found for both players and returns the $(\hat{\mathbf{x}}, \hat{y})$ of the last iteration (lines 9-13).

If the DO-NS procedure considers mixed strategies when solving a restricted game, this algorithm serves as an exact algorithm [20], and the mixed-strategy profile returned by the DO-NS algorithm is the Nash equilibrium of the game. The core approach of the DO-NS algorithm is to solve the restricted game and find the best response strategy. The latter is introduced separately.

A. CoreLP

During the kth iteration of the DO-NS algorithm, $U^k = [u(s_R, s_B)]_{|S_R^k| \times |S_B^k|}$ is obtained by calculating the corresponding utility of each strategy profile in S^k; then, the mixed-strategy Nash equilibrium is obtained by solving G^k. Because G^k is a smaller restricted game, formulation (10) in

Section II-B can be transformed into a dual linear programming model for solving [30], such as LP1 and LP2.

$$\text{LP1} \quad \max\{v\} \tag{11}$$

$$\text{s.t.} \sum_{s_R \in S_R^k} u(s_R, s_B)\, x_{s_R} \geq v, \quad \forall s_B \in S_B^k \tag{12}$$

$$\sum_{s_R \in S_R^k} x_{s_R} = 1 \tag{13}$$

$$x_{s_R} \geq 0, \forall s_R \in S_R^k \tag{14}$$

$$\text{LP2} \quad \min\{v\} \tag{15}$$

$$s.t. \sum_{s_B \in S_B^k} u(s_R, s_B)\, y_{s_B} \leq v, \quad \forall s_R \in S_R^k \tag{16}$$

$$\sum_{s_B \in S_B^k} y_{s_B} = 1 \tag{17}$$

$$y_{s_B} \geq 0, \quad \forall s_B \in S_B^k \tag{18}$$

The optimal solutions $\hat{\mathbf{x}}$ and $\hat{y}$ obtained by solving LP1 and LP2 are the mixed-strategy Nash equilibrium, which is solved by CPLEX. In the 0th iteration of the DO-NS algorithm, this paper uses a uniform distribution mechanism to select the pure strategies of both sides as the strategy set of a restricted game. For instance, there are $|M|$ UAVs that should choose cubes $\left\lceil g/|M| \right\rceil$, $\left\lceil g/|M| \right\rceil \times 2, \ldots, \left\lceil g/|M| \right\rceil \times |M|$ as the occupancy positions among g cubes.

B. Red Oracle/Blue Oracle

Red Oracle or Blue Oracle finds the best occupancy strategy of R or B given a current occupancy strategy of the opponent for the kth iteration. Given a restricted game G^k and its solution $(\hat{\mathbf{x}}, \hat{y})$, the best response strategy of R is the pure strategy $\bar{s}_R \in S_R$ that satisfies $\sum_{s_B \in S_B^k} u(\bar{s}_R, s_B)\, \hat{y}_{s_B} \geq \sum_{s_B \in S_B^k} u(s'_R, s_B)\, \hat{y}_{s_B}$ for all $s'_R \in S_R$, and the best response strategy of B is the pure strategy $\bar{s}_B \in S_B$ that satisfies $\sum_{s_R \in S_R^k} u(s_R, \bar{s}_B)\, \hat{x}_{s_R} \leq \sum_{s_R \in S_R^k} u(s_R, s'_B)\, \hat{x}_{s_R}$ for all $s'_B \in S_B$.

Through the above analysis, the best response strategies of R and B can be formulated as BR1 and BR2, respectively.

$$\text{BR1} \quad \text{maximize}_{\mathbf{d_R}} \sum_{s_B \in S_B^k} u(\mathbf{d_R}, s_B)\, \hat{y}_{s_B} \tag{19}$$

$$s.t. \sum_{e \in C_R} d_R^e = |M| \tag{20}$$

$$d_R^e \in \{0, 1\} \tag{21}$$

$$\text{BR2} \quad \text{minimize}_{\mathbf{d_B}} \sum_{s_R \in S_R^k} u(s_R, \mathbf{d_B})\, \hat{x}_{s_R} \tag{22}$$

$$s.t. \sum_{e \in C_B} d_B^q = |N| \tag{23}$$

$$d_B^q \in \{0, 1\} \tag{24}$$

BR1 is a bilevel program because the objective function $\sum_{s_B \in S_B^k} u(\mathbf{d_R}, s_B)\, \hat{y}_{s_B}$ needs to solve two inner maximization problems on the decision variables $\mathbf{a}$ and $\mathbf{b}$ for a given strategy $\mathbf{d_R}$. Similarly, BR2 is also a bilevel program. The optimal solution of the above two-level optimization problem can be solved by an enumeration method. However, when the number of occupancy strategies is large, this enumeration will take a large amount of time. For R, we utilize a neighborhood search (NS) to find an improving strategy $\bar{s}_R \in S_R$ and a strategy that satisfies $\sum_{s_B \in S_B^k} u(\bar{s}_R, s_B)\, \hat{y}_{s_B} \geq \sum_{s_B \in S_B^k} u(s'_R, s_B)\, \hat{y}_{s_B}$ for $\forall s'_R \in S_R^k$ but not for $\forall s'_R \in S_R$. The NS can effectively reduce the search strategy space. Taking the best response strategy for solving R as an example, the pseudocode of the algorithm is shown in algorithm 2, and the solution process of the best response strategy for B is the same.

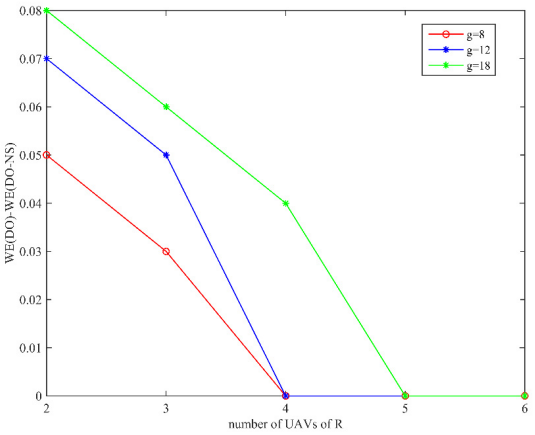


FIGURE 5. The difference between the worst expected utilities of the solution obtained by DO algorithm and DO-NS algorithm of R when $|N| = 4$.

Algorithm 2 Red Oracle Algorithm

1. $\bar{s}_R \leftarrow \underset{s_R \in S_R^k}{\text{maximize}}\, \hat{x}_{s_R}$
2. $\bar{u} = -\infty$
3. repeat
4. $\quad Ne \leftarrow neighbors(\bar{s}_R)$
5. $\quad$ for i =1 to $|Ne|$ do
6. $\quad\quad$ if $\sum_{s_B \in S_B^k} u(Ne(i), s_B)\, \hat{y}_{s_B} \geq \bar{u}$
7. $\quad\quad\quad \bar{s}_R \leftarrow Ne(i)$
8. $\quad\quad\quad \bar{u} \leftarrow \sum_{s_B \in S_B^k} u(Ne(i), s_B)\, \hat{y}_{s_B}$
9. $\quad\quad$ end if
10. $\quad$ end for
11. until $\bar{s}_R = Ne(1)$
12. return $\bar{s}_R$

We use the pure strategy with the maximum probability in equilibrium strategy $\hat{\mathbf{x}} = (\hat{x}_{s_R})^T_{s_R \in S_R^k}$ (i.e., $\bar{s}_R = \max_{s_R \in S_R^k} \hat{x}_{s_R}$) as the current strategy $\bar{s}_R$ (line 1) and apply a 2-opt exchange [29] to generate all neighbor strategies of $\bar{s}_R$; the set of neighbor strategies is denoted by Ne (line 4), where $Ne(i)$ is the ith strategy of Ne and $\bar{s}_R$ is the first element of Ne, that is, $\bar{s}_R = N(1)$. Then, the strategy with the maximum expected utility in Ne is set as the new $\bar{s}_R$ (lines 5-10). If $\bar{s}_R$ is $N(1)$, then return $\bar{s}_R$ (lines 11-12). Otherwise, enter the next iteration, regenerate the neighbor strategies of $\bar{s}_R$, and find the new $\bar{s}_R$. The strategy $\bar{s}_R$ returned by the algorithm is the best response strategy.

TABLE 1. Problem instances, computational time and worst expected utility.

Instance	g	$\|M\|$	$\|N\|$	$\|S\|$	Computational time (sec)		Worst expected utility	
					DO	DO-NS	DO	DO-NS
1	8	2	4	1960	9.33	10.97	-1.49	-1.49
2	8	3	4	3920	57.59	64.65	-0.77	-0.80
3	8	4	4	4900	57.29	61.35	0.00	0.00
4	8	5	4	3920	75.02	85.65	0.49	0.49
5	8	6	4	1960	58.46	81.67	0.97	0.97
6	12	2	4	32670	123.69	74.96	-1.53	-1.60
7	12	3	4	108900	365.03	124.58	-0.75	-0.80
8	12	4	4	245025	482.81	139.40	0.00	0.00
9	12	5	4	392040	1015.09	449.98	0.75	0.75
10	12	6	4	457380	648.40	197.55	1.50	1.50
11	18	2	4	468180	383.63	193.40	-1.53	-1.61
12	18	3	4	2496960	2329.04	208.05	-0.74	-0.80
13	18	4	4	9363600	3187.15	376.47	0.00	0.04
14	18	5	4	26218080	8835.19	1257.53	0.76	0.76
15	18	6	4	56805840	12916.69	667.63	1.52	1.52
16	32	2	4	17836160	2818.70	370.29	-1.52	-1.67
17	32	3	4	178361600	-	730.45	-	0.84
18	32	4	4	1293121600	-	876.87	-	0.00
19	32	5	4	7241480960	-	474.13	-	0.71
20	32	6	4	32586664320	-	549.56	-	1.51

In algorithm 2, the 2-opt exchange is that there is a UAV that moves from any cube $c_R^i \in C_R$ to some cubes $c_R^j \in C_R, i \neq j$ that are not occupied. For example, in an occupancy area that is divided into four cubes, that is, $C_R = \{c_R^1, c_R^2, c_R^3, c_R^4\}$, if a given initial occupancy strategy is [0, 1, 0, 1], where two UAVs occupy cubes 2 and 4, respectively, the neighborhood of solutions generated from such an occupancy strategy via a possible 2-opt exchange is

$$\textbf{neighbors}\,([0,1,0,1]) = \left\{ \begin{array}{l} [0,1,0,1] \\ [1,0,0,1] \\ [0,0,1,1] \\ [1,1,0,0] \\ [0,1,1,0] \end{array} \right\}$$

wherein each row corresponds to a neighbor strategy of the input strategy.

IV. EXPERIMENT AND ANALYSIS

To test the performance of the proposed method, two aspects of experimental analysis are carried out. Section IV-A analyzes the scalability of the DO-NS algorithm, and Section IV-B discusses the quality of solutions obtained by the DO-NS algorithm. We use CPLEX (12.7.1) to solve all linear programs. The DO algorithm procedure, the DO-NS algorithm procedure and the call to CPLEX were coded in Microsoft Visual Studio Ultimate 2013. All experiments were run on a 64-bit PC with 16.0 GB RAM and a 3.60 GHz processor. In the experiments, the length, width and height of the occupancy area of R and B are 30 km, 30 km and 10 km, respectively. The minimum and maximum firing ranges of the weapons carried by the UAVs are 30 km and 55 km, and the best air combat altitude is 5 km. The other parameters w_1 and w_2 are 0.6 and 0.4, respectively.

A. SCALABILITY ANALYSIS

This section analyzes the scalability of the DO-NS algorithm by comparing the time requirements and the worst expected utilities of the solutions obtained by the DO algorithm and DO- NS algorithm in different problem instances. According to differences in g, $|M|$ and $|N|$, we design 20 problem instances, as shown in Table 1. Here, g represents the number

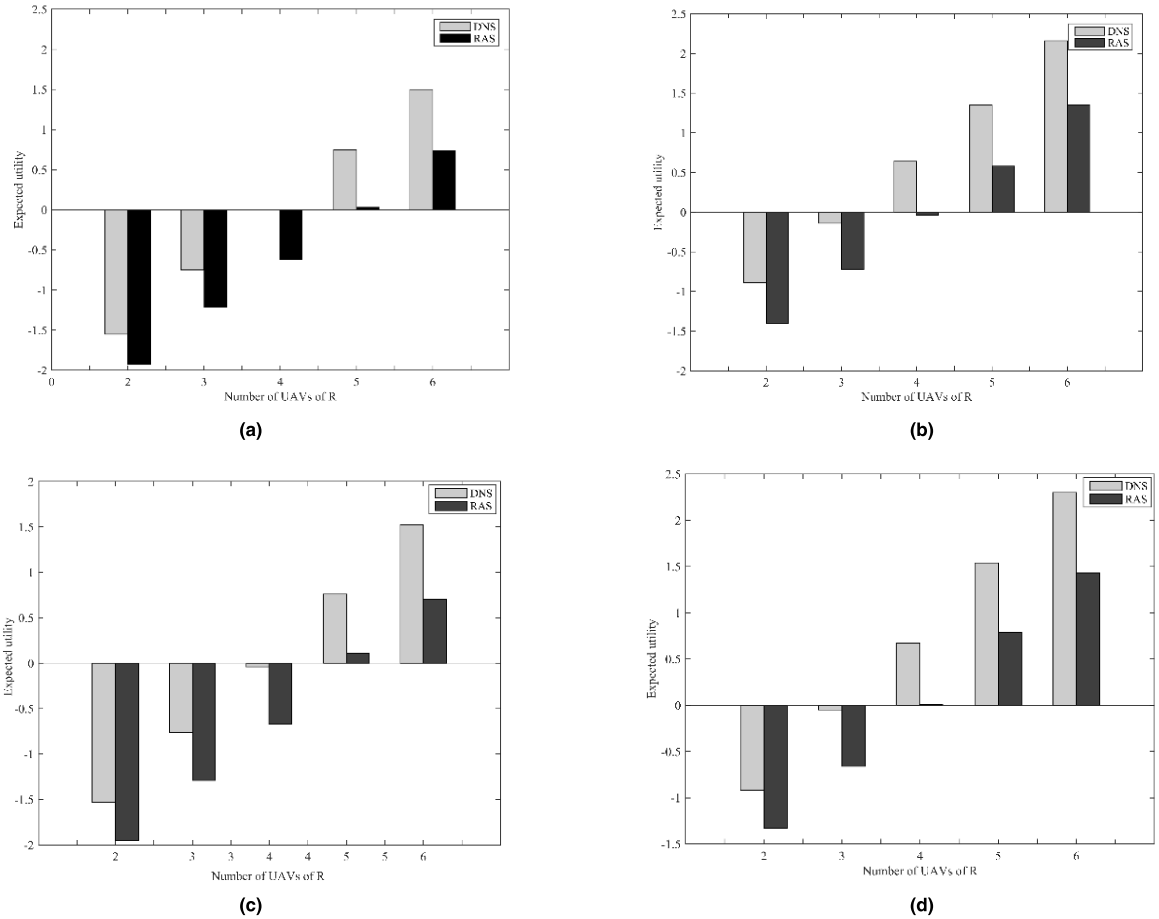


FIGURE 6. Solution quality when $|N| = 4$. (a) B performing DNS when $g = 12$; (b) B performing RAS when $g = 12$; (c) B performing DNS when $g = 18$; (d) B performing RAS when $g = 18$.

of cubes in the occupancy area and has four different values. The first value divides the length, width and height of the occupancy area into two parts, forming 8 cubes, and $g = 8$. The second value divides the length of the occupancy area into three parts and the width and height into two parts, forming 12 cubes, and $g = 12$. The third value divides the length and width of the occupancy area into three parts and the height into two parts, forming 18 cubes, and $g = 18$. The fourth value divides the length and width of the occupancy area into four parts and the height into two parts, forming 32 cubes, and $g = 32$. $|M|$ and $|N|$ represent the numbers of UAVs of R and B, respectively, and $|S|$ represents the number of combinations of occupancy strategies for both sides of the confrontation. In addition, all the experimental results are averaged 15 times unless otherwise specified.

In Table 1, the sixth and seventh columns show the computational times of the DO algorithm and DO-NS algorithm for 20 problem instances, respectively, and the eighth and ninth columns show the worst expected utilities of R when R performs the solutions obtained by the DO algorithm and DO-NS algorithm on the 20 problem instances, respectively (i.e., the worst expected utility of R is the utility obtained by R when R performs the equilibrium strategy and B performs the best response strategy to the equilibrium strategy of R). The "-" in the sixth column indicates that the calculation time exceeds 20,000 seconds, so we do not obtain the Nash equilibrium solution.

First, we analyze the computational time of the algorithm. As shown in Table 1, when $|S|$ is less than 10000, there is little difference between the DO-NS algorithm and the DO algorithm in computational time. However, when $|S|$ increases above 10000, the DO-NS algorithm shows better performance in the computational time. This result is because when the scale of the strategies is small, the neighbor strategies generated by the DO-NS algorithm in each iteration will cover the whole strategy space, so the computational times have little difference from those of the DO algorithm. When the scale of the strategies is large, the DO-NS algorithm can greatly improve the search efficiency of the best response strategy through effective neighbor strategy generation and iterative search. For the problem instance of $|M| = |N| = 4$ and $g = 18$, the computational time of the DO algorithm is more than 3000 seconds, while the DO-NS algorithm needs only 300 seconds. For the problem instance of $|M| = 6$, $|N| = 4$ and $g = 18$, the computational time of the DO algorithm is approximately 13000 seconds, while the DO-NS

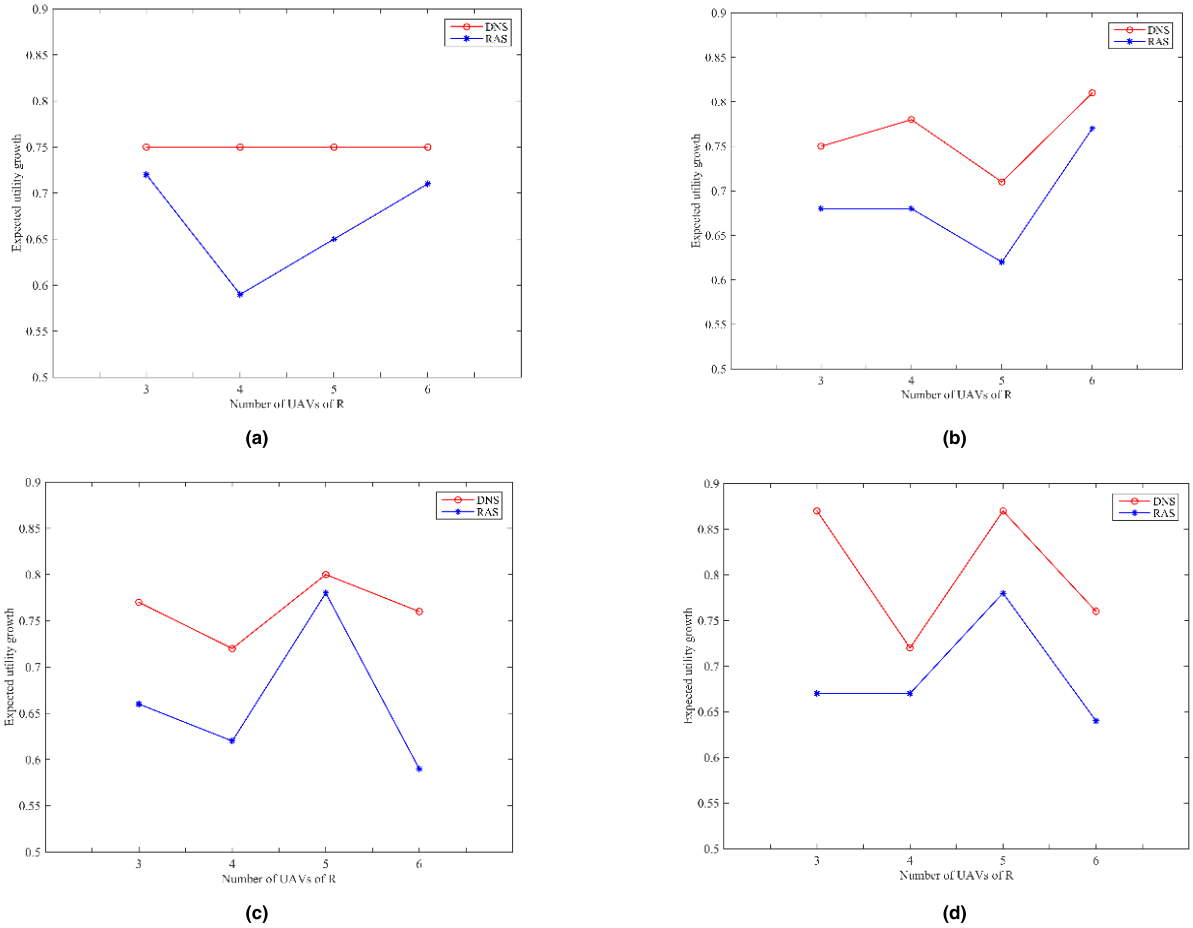


FIGURE 7. Expected utility growth when the number of UAVs of R increases by 1 and $|N| = 4$. (a) B performing DNS when $g = 12$; (b) B performing RAS when $g = 12$; (c) B performing DNS when $g = 18$; (d) B performing RAS when $g = 18$.

algorithm can obtain the equilibrium strategy in approximately 600 seconds.

Second, we analyze the expected utility of the solutions obtained by the algorithm in the worst case. As shown in the last two columns in Table 1, in the first fifteen problem instances, there are nine problem instances in which the solutions obtained by the DO-NS algorithm are the same as those obtained by the DO algorithm. The worst expected utility of the solution obtained by the DO-NS algorithm is slightly lower than that obtained by the DO algorithm in the other seven problem instances. When g is invariant, as the scale of the game strategies increases, the differences between these values decrease. For the above problem instances, we compare the differences in the worst expected utility between the DO algorithm and the DO-NS algorithm, as shown in Fig. 5. In the same division of the occupancy areas, the more UAVs there are, the smaller the difference between the worst expected utility of the DO-NS algorithm and that of the DO algorithm. The above analysis shows that the DO-NS algorithm can save a large amount of computational time on the premise of less utility loss, thereby effectively extend the application of large-scale strategy problem examples.

B. SOLUTION QUALITY

In this section, we compare the quality of the solution obtained by the DO-NS algorithm with that of a baseline solution. The solution obtained by the DO-NS algorithm is called DNS, which is a mixed-strategy where each pure strategy s_R is selected with probability $p(s_R) = x^*_{s_R}$, and the UAV occupies the position according to the actually selected strategy s_R. The baseline solution is a random solution called RAS, which is a mixed-strategy where each pure strategy s_R is selected with the same probability $p(s_R) = 1/\tau_R$, where the parameter $\tau_R$is the number of strategies of R.

In the experiments, the number of UAVs of B remains unchanged, and the number of UAVs of R is 2, 3, 4, 5 or 6. For occupancy areas divided into 12 cubes and 18 cubes, we compare the expected utility of B performing DNS and RAS with that of R performing DNS and RAS, as shown in Fig. 6. The expected utilities are averaged 100 times.

In Fig. 6, the x-axis represents the number of UAVs of R, and the y-axis represents the expected utility of R. Fig. 6 shows that, first, no matter what solution B executes, the expected utility of R when executing DNS is greater than that of RAS, and this finding shows the effectiveness of the Nash equilibrium. Second, when the number of UAVs of B

remains unchanged, the expected utility of R increases with the number of UAVs of R. When the numbers of UAVs on both sides are the same, the expected utility of the same solution is the same. For example, when the number of UAVs on both sides is 4, the utility of DNS or RAS is 0, which shows the importance of resources. In addition, when the UAV resources increase, the expected utility growth of DNS is greater than that of RAS, as shown in Fig. 7, where the x-axis represents the number of UAV resources of R increased by 1 and the y-axis represents the expected utility growth of the UAV resource increased by 1. For instance, as shown Fig. 7 (d), $g = 18$, $|N| = 4$, B executes RAS, and $|M| = 6$, the expected utility of R when executing DNS is 0.76 higher than in that when $|M| = 5$, but the expected utility of R when executing RAS is higher by only 0.64. That is, when UAV the resources increase, DNS can bring a higher expected utility. In addition, because RAS is a random strategy and DNS is a mixed-strategy Nash equilibrium that brings fluctuations in utility, the expected utility growth present a zig-zag trend. Through the above experiments and analysis, the quality of the Nash equilibrium solution in the multi-UAV cooperative occupancy decision-making problem is illustrated.

V. CONCLUSION AND FUTURE WORK

In this paper, by studying the influence of different occupancy strategies of UAVs on the subsequent cooperative target assignment, a new cooperative occupancy decision-making problem is proposed for multi-UAV in BVR air combat. This problem is a two-stage decision-making problem; that is, the decision maker first decides the occupancy position of each UAV, then decides the target of each UAV based on the occupancy of both the UAVs and targets. By analyzing the characteristics of the problem, we model it as a zero-sum matrix game model and design a DO-NS algorithm to find the mixed-strategy Nash equilibrium of the model. A large number of experimental results show that compared with the DO algorithm, the DO-NS algorithm has high efficiency in solving large-scale strategy problems. The worst expected utility of the solution obtained in a confrontation is slightly lower than that obtained by the DO algorithm. This result shows that the DO-NS algorithm can save much time with less utility loss. In addition, compared with a random solution, the solution obtained by the DO-NS algorithm can achieve higher efficiency, which shows the effectiveness of the Nash equilibrium in this problem. In future work, we have an interest in considering the multi-UAV cooperative occupancy decision-making problem with uncertainties, such as the statue uncertainty and performance uncertainty of each UAV, and in further studying methods to solve these uncertainties to improve the applicability of the proposed method.

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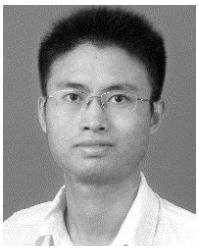
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